MILITARY SPECIFICATION

# RANGES, ELECTRIC, COMMERCIAL (WITH OR WITHOUT BROILER)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers commercial, heavy-duty, electric ranges with or without broiler.

1.2 Classification. Ranges covered by this specification shall be of the following types, styles and grades as specified (see 6.2):

| | | |
|---|---|---|
| Type I | - | Hot top. |
| Type II | - | Griddle top. |
| Type III | - | Griddle and hot top. |
| Style 1 | - | With back shelf broiler. |
| Style 2 | - | Without back shelf broiler. |
| Grade A | - | Stainless steel. |
| Grade B | - | Carbon steel with manufacturer's commercial finish. |

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-R-11337 - Ranges; Packaging and Packing of.

FSC 7310

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of specifications, standards and regulations required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Underwriters' Laboratories, Inc. (UL)

Standard No. 197 - Commercial Electric Cooking Appliances.

Standard No. 20 - Switches, Snap.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, L.I., NY 11746; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

National Sanitation Foundation (NSF)

Standard No. 4 - Commercial Cooking and Hot Food Storage Equipment.

Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

American National Standards Institute (ANSI)

Standard B46.1 - Surface Texture (Surface Roughness, Waviness and Lay).

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)



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American Society for Testing and Materials (ASTM)

Designation A167-70 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip.

Designation A176-71 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.3 and 6.3).

3.2 Standard product. The electric commercial ranges shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

* 3.3 Codes and standards. The electric commercial ranges shall comply with the applicable requirements of UL Standard Nos. 197 and 20 and NSF Standard No. 4.

3.3.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the electric commercial ranges he proposes to furnish under this specification meet the applicable requirements of UL Standard Nos. 197 and 20 and NSF Standard No. 4.

3.3.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the electric commercial ranges have been tested and conform to UL Standard Nos. 197 and 20.



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3.3.1.2 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished electric commercial range, or

(2) A certification for the electric commercial range issued by NSF under their special one-time contract evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the commercial electric ranges are being procured, indicating that the commercial electric ranges have been tested and conform to NSF Standard No. 4.

* 3.4 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be the quality normally used by the manufacturer in his standard commercial electric range provided the completed item complies with all provisions of this specification.

3.4.1 Stainless steel. Stainless steel shall conform to any type 300 or 400 steel specified in ASTM A167 or A176, respectively.

3.4.2 Insulation. Insulation shall be board or blanket type thoroughly dry mineral wool or fiberglass. Loose fill insulation shall not be used.

3.5 Design. Ranges shall consist essentially of an oven section surmounted by a hot top (type I), griddle top (type II), or dual purpose griddle and hot top (type III) heating surface, heating elements, and controls. Types and styles of ranges shall be as indicated in 1.2. The ends of the ranges shall be designed so that they can be banked together in battery alignment with no space between range tops. There shall be no permanent distortion or damage to any part of the completely assembled range when tested as specified in 4.6.2.

3.5.1 Electrical. Unless otherwise specified (see 6.2), ranges shall be wired for operation on a nominal 208-volt, 3-phase, 60-hertz (hz) alternating current power system with a conversion jumper for 208-volt, single-phase, 60-hz operation. Unidentified leads to terminals shall be stamped or branded to correspond with similar markings on terminals. All electric wiring shall be tension-free and clamped between corrosion-resisting steel terminal parts. All heater current-carrying wires installed in a heated zone that will exceed 400 degrees Fahrenheit (F), or connected to heating



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element terminals where the wire connection temperature exceeds 400 F shall be nickel wire or nickel-clad copper wire not smaller than No. 12 American Wire Gage (AWG) with heavy, silicon impregnated asbestos insulation which shall withstand a 2500-volt 60-hz, alternating current potential at 550 F for one minute without breaking down when tested as specified in 4.6.3. Control circu wires of the above description shall be not smaller than No. 18 AWG. Wire to be brazed or welded shall be manganese-nickel not smaller than No. 12 AWG. Al wire carrying heater current not in a heated zone, shall be tinned solid coppe not smaller than No. 12 AWG with heavy waterproof and flameproof asbestos insulation.

3.5.2 Indicator lights. Each thermostat shall be furnished with an indicator light which shall be illuminated when the heating element which it controls is energized.

* 3.5.3 Wattages. For operational voltages of 208, 230/240, or 460/480, range heating elements shall have wattages as specified in table I. For operational voltages of 220 or 440, a minus tolerance of 10 percent is applicable to minimum wattages specified for "HIGH" position in table I (see 4.6.9).

TABLE I. Heating element wattages 1/

| Range classification | Switch position | | | |
|---|---|---|---|---|
| | High | | Medium | Low |
| | Min. | Max. | | |
| Type I, style 2 | | | | |
| Three top plates | 4,750 each | 5,565 each | 1/2 of high | 1/2 of medium |
| Oven (2 elements) | 2,850 each | 3,625 each | 1/2 of high | 1/2 of medium |
| Type II, style 2 | | | | |
| One full width plate | 14,250 total | 17,010 total | N/A | N/A |
| Oven (2 elements) | 2,850 each | 3,625 each | 1/2 of high | 1/2 of medium |
| Type III, style 2 | | | | |
| Three top plates | 4,750 each | 5,565 each | 1/2 of high | 1/2 of medium |
| Oven (2 elements) | 2,850 each | 3,625 each | 1/2 of high | 1/2 of medium |
| Broiler (style 1) | 4,940 each | 6,300 each | 1/2 of high | 1/2 of medium |

1/ Above wattages are for steady state conditions.



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* 3.5.4 Thermostats. Oven thermostats shall be calibrated at 400 F, griddle-plate thermostats at 350 F, and top plate thermostats at 550 F. Oven and top plate thermostats set at their calibration points shall maintain this average temperature, plus or minus 10 F, when tested as specified in 4.6.1. The oven temperature amplitude for cycle (difference between the maximum and minimum temperatures) shall not exceed 40 F and the top plate temperature amplitude per cycle shall not exceed 70 F when tested as specified in 4.6.1. All thermostats shall be marked in increments of 25 F.

3.6 Construction.

3.6.1 Body. The range body shall be either welded-frame type or inner-and-outer-shell type of construction with joints secured to form a structure that will support all parts. Required sheet metal thicknesses are listed in table II. Overall range dimensions and oven dimensions are shown in table III.

TABLE II. Nominal sheet metal thickness (inches) 1/

| Component | Grade A | Grade B | |
|---|---|---|---|
| | | Inner and outer shell construction | Welded frame construction |
| Side and front exterior panels | 0.0500 | 0.0478 | 0.0359 |
| Back exterior panel | 0.0375 | 0.0359 | 0.0359 |
| Bottom exterior panel | 0.0359 | 0.0359 | 0.0359 |
| Oven inner lining (top, bottom, back) | 0.0359 | 0.0359 | 0.0359 |
| Oven inner lining (sides) 2/ | 0.0598 2/ | 0.0598 2/ | 0.0598 2/ |
| Door inner lining | 0.035 | 0.035 | 0.035 |
| Oven deck | 0.0598 | 0.0598 | 0.0598 |
| Drip pan | 0.0359 | 0.0359 | 0.0359 |
| Back shelf (broiler) | 0.0375 | 0.0359 | 0.0359 |
| Drip pan (broiler) | 0.0375 | 0.0359 | 0.0359 |

1/ Thicknesses may be greater.



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2/ Oven side linings, if reinforced, may be the same gage as top, bottor and back linings.

TABLE III. Range dimensions, inches

| | Width | Depth | Height |
|---|---|---|---|
| Oven interior (min) | 22-1/4 | 26-1/2 | 12 |
| Range, overall (min) | 36 | 38 1/ | 36 (see 3.6.2) |

1/ Includes front overhang.

3.6.1.1 Grade A range. The exterior front, side panels, and back panel of the grade A range shall be stainless steel as specified in 3.4.1

3.6.1.2 Grade B range. The exterior front, side panels, and back pane of the grade B range shall be carbon steel.

3.6.2 Legs. The ranges shall have an adjustable leg attached to each bottom corner of the body to provide an adjustment of plus or minus 1-inch to the 36-inch height specified (see table III). The adjustable legs for grade A ranges shall be stainless steel as specified in 3.4.1.

3.6.3 Top heating surface. The top heating surface shall be made up of hot plates of the type and size specified in 3.6.3.1, 3.6.3.2 and 3.6.3.3. Hot plates shall be made of cast iron, cast steel, steel boilerplate or any type of hot-rolled steel. Cast iron and cast steel hot plates shall have flanges on the underside near all edges. Steel boilerplate, cast steel, and hot-rolled steel hot plates shall have a minimum thickness of 7/16 inch. Cast iron hot plates shall be a minimum of 3/8 inch thick. The top surface shall be flat and not coarser than 125 micro inches in accordance with ANSI B46.1.

3.6.3.1 Type I (hot top). Type I ranges shall consist of three hot plates as specified in 3.6.3. Each hot plate shall be 12 inches wide, plus or minus 1/8 inch, by 24 inches long, plus or minus 1/8 inch, and shall be installed with the long dimension from front to rear of the range. Each of the three plates shall be separately controlled by one or more 3-heat indicating switches marked "LOW", "MEDIUM", "HIGH" and "OFF" or by an automatic, adjustable thermostat specified in 3.5.4 having an "OFF" position. When a thermostat is used, it shall have a minimum range



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of from 250 F to 850 F. When 3-heat indicating switches are used, the medium and low heat position shall be one-half and one-fourth respectively, of high heat (see 3.5.3). Preheat time shall be sufficient to produce a surface temperature of the hot top heating surface of not less than 500 F in not more than 20 minutes when tested as specified in 4.6.4. The heat distribution shall be such that the temperature, measured simultaneously at 12 points, shall not vary by more than plus or minus 14 percent of the average reading when tested as specified in 4.6.5.

3.6.3.2 Type II (griddle top). Type II ranges shall consist of a single, full-width griddle-plate 36 inches wide, plus or minus 1/2 inch, by 24 inches long, plus or minus 1/8 inch, with side and rear back-splash guards. The griddle-plate shall be controlled by not less than four thermostats as specified in 3.5.4, each having an "OFF" position. The thermostats shall be capable of controlling the top surface heating temperature of the plate from 250 F to 450 F. The griddle-plate shall be at rated voltage and shall be sufficient to produce a temperature over the entire top heating surface of not less than 400 F in a preheat time of not more than 25 minutes when tested as specified in 4.6.4. The heat distribution shall be such that the temperatures, measured simultaneously at 12 points shall not vary by more than plus or minus 14 percent of the average reading when tested as specified in 4.6.5. The griddle-plate shall be provided with a grease trough and a minimum of two drains which shall empty into a grease receptacle, removable from the front of the range.

3.6.3.3 Type III (griddle and hot top). Type III range shall consist of three dual purpose top surface heating hot plates which shall be suitable for either hot top or griddle top cooking operations. The three hot plates shall be clamped together into one assembly, 36 inches wide, plus or minus 1/2 inch, by 24 inches long, plus or minus 1/8 inch, and shall be installed with the 24-inch dimension from front to rear of the range. A heat insulating spacer shall be installed between adjacent hot plates to minimize heat transfer. Each hot plate shall be controlled by one 3-heat indicating switch and automatic adjustable thermostat as specified in 3.5.4, or by an automatic adjustable thermostat having an "OFF" position. The thermostat shall have a minimum range from 250 F to 850 F. When 3-heat indicating switches are used, the medium and low heat position shall be one-half and one-fourth, respectively, of high heat position (see 3.5.3). The 3-heat indicating switches shall be marked "LOW" "MEDIUM", "HIGH", and "OFF" as applicable. Preheat time shall be sufficient to produce a top surface heating temperature of not less than 500 F, in not more than 20 minutes when tested as specified in 4.6.4. The heat distribution shall be such that the temperature, measured simultaneously at 12 points, shall



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not vary by more than plus or minus 14 percent of the average reading when tested as specified in 4.6.5. The hot plates shall be provided with a grease trough having a minimum of two drains which shall empty into a grease receptacle removable from the front of the range.

3.6.4 Ovens. The ovens for grade A and B ranges shall be constructed so that liquids will not drain into the oven bottom. The ovens shall be insulated against heat loss with one of the materials specified in 3.4.2. Insulation shall be installed so that it fills all spaces and does not settle and leave portions of the oven unprotected. The ovens shall be provided with a dampered vent, located at the back and extending to the top of the range. The damper shall be controlled by a lever from the front of the range. A minimum of one slide rack and supports shall be provided, positioned to divide the oven into at least two equal compartments horizontally.

3.6.4.1 Oven door. The oven door shall be hinged at the bottom, spring counter-balanced, and when fully opened shall lie and remain level and even with the oven deck. The door opening shall be the full width and height of the oven interior and the door shall be capable of supporting a dead weight load of 150 pounds placed over the outer edge when the door is in the open position, without breakage or permanent distortion when tested as specified in 4.6.6. Means shall be provided for lubricating door hinges, if not permanently lubricated type.

3.6.4.2 Oven deck. A deck, having dimensions commensurate with the oven bottom and with raised flanges at the back and both sides, shall be provided over the lower heating element. The deck shall be constructed and reinforced to prevent warping, and shall be removable through the oven door without removing the door or any part of the range front.

3.6.4.3 Oven heat loss. The oven heat loss shall not be greater than 55 watts per square foot of inside oven surface area while maintaining a thermostatically controlled internal temperature of 350 F when tested as specified in 4.6.7.

3.6.4.4 Oven heating elements. The oven heating elements shall be the inclosed type having the resistance wire encased in a sealed metal tubing or sheathing. The elements shall be removable from the front of the range and shall provide an even distribution of heat throughout the oven. There shall be two heating sections, one located at the top and one at the bottom of the oven chamber. Each oven heating section shall be controlled by a 3-heat indicating switch and automatic, adjustable thermostat calibrated from 200 F to 500 F. In lieu of the 3-heat indicating switches and the automatic, adjustable thermostat specified in 3.5.4, the supplier may



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furnish a thermostat having an "OFF" position with two separate sets of contacts to provide separate control of top and bottom elements. The actuating element of the oven thermostat shall be protected by a perforated or expanded metal shield. The 3-heat indicating switches shall be marked "LOW", "MEDIUM", "HIGH" and "OFF".

3.6.4.5 Oven preheat time. When operated at rated input voltage and the temperature control at the high heat position, the oven temperature shall rise to 400 F in not more than 20 minutes with the oven empty and the oven door closed when tested as specified in 4.6.8.

3.6.5 Broiler. Style 1 ranges shall be provided with a broiler attached to the back of the top heating surface of the range. The broiler shall be mounted in a back shelf. The back shelf shall be fabricated of stainless steel specified in 3.4.1 for grade A ranges and of carbon steel for grade B ranges. The height of the back shelf above the top heating surface shall not exceed 37-1/2 inches, the width shall be the same as the range and depth shall not exceed 18 inches excluding the control handle. All parts of the broiler on the grade A range and back shelf except the grid, but including broiler inner linings, hardware, fastening devices, broiler deck, drip pan, and exterior surfaces shall be fabricated of stainless steel specified in 3.4.1. All exposed interior and exterior surfaces of the grade A range broiler and back shelf shall have a minimum of No. 3 standard finish. All interior and exterior surfaces of the grade B range broiler and back shelf shall have the manufacturer's standard finish. The broiler grid for each grade range shall have a surface area of at least 288 square inches. The grids shall be removable and shall be adjustable vertically.

3.6.5.1 Broiler heating element. The broiler heating element shall be the inclosed type having a resistance wire encased in a sealed metal tubing or sheathing. The broiler heating element shall be controlled by an automatic, adjustable thermostat with an "OFF" position or by a 3-heat indicating switch, clearly marked "HIGH", "MEDIUM", "LOW", and "OFF". The energy input at rated voltage shall be as indicated in table I.

3.6.5.2 Broiler drip pan. Broilers shall be provided with a removable drip pan which shall fit at the bottom of the broiling compartment to catch all broiler grid spillage. The grade A range broiler drip pan shall be fabricated of stainless steel specified in 3.4.1. The grade B range broiler drip pan shall be fabricated of carbon steel. The grade A range drip pan shall have a minimum No. 3 standard finish on external surfaces. The grade B range drip pan shall have the manufacturer's standard finish. Drip pans shall be removable from the front.



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3.7 Marking.

3.7.1 Identification plate. Identification marking shall be etched or stamped on a stainless steel or aluminum plate in accordance with MIL-STD-130. In addition, electrical characteristics and information required by UL Standard No. 197 shall also appear on the plate.

3.7.2 User's instruction plate. Each range shall be equipped with a user's instruction plate, fabricated of stainless steel or aluminum sheet having a minimum thickness of 0.025 inch. The plate shall be of sufficient overall size to contain the following information, etched, embossed, or die stamped in legible characters:

RANGE-ELECTRIC -

USER'S INSTRUCTIONS

1. Preheat oven by turning both heating elements to high heat for a short period. Allow 20 minutes to heat oven to 400 F, then turn off upper heating element; turn lower heating element on medium heat and set the thermostat to regulate temperature.
2. Allow 25 minutes for heating top heating surfaces.
3. When top heating surface is controlled by a thermostat, set thermostat for heat desired (350 F to 450 F for average use) to obtain uniform temperature while cooking.
4. Avoid damaging the heating elements or spilling food on them.
5. If any liquid spills on oven baffle plate or deck, wipe it off as soon as possible.
6. When range is cool, clean exterior with damp cloth and wipe or brush out oven.
7. Be sure switches are turned off when range is not in use.

NOTE 1. The instructions shall be in characters not less than 3/32 inch high except the words "RANGE-ELECTRIC" shall be in characters not less than 3/16 inch high; the words "USER'S INSTRUCTIONS," shall be in characters not less than 1/8 inch high.

NOTE 2. The user's instruction plate shall be permanently attached to the outside front of the oven door.



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3.8 Finish. The range shall be clean with no foreign matter imbedded in the finish. The finish shall be free from scratches, gouges, pits, chips, grit, rust, or scale. Rough or sharp edges, burrs, and slivers shall be removed.

3.8.1 Grade A range. Stainless steel exterior surfaces (front, sides and back) of the grade A range shall have a No. 3 standard finish, minimum. The range bottom and the oven interior, including the inner surface of the door, shall have the manufacturer's standard finish. The top heating surface shall be finished as specified in 3.6.3.

3.8.2 Grade B range. All exterior surfaces of the grade B range shall have the manufacturer's standard finish. Oven interior surfaces, including the inner surface of the door, shall have the manufacturer's standard finish. The top heating surface shall be finished as specified in 3.6.3.

3.9 Repair and maintenance. Provisions shall be made to allow for ready adjustment, service, and replacement of heating units, top heating surfaces, and thermostats, without disturbing or disassembly of any major components or equipment. In addition, there shall be free access to inspect, service, and adjust any component or equipment as provided for by this specification or as indicated by the component or equipment manufacturer, without requiring disassembly of any other component or equipment on the end item.

3.10 Workmanship. The finished range shall be clean, well made, suitable for the intended use. All components shall be carefully assembled, properly adjusted and aligned. All fasteners shall be drawn up tight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:



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1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article. When a first article is required, the range shall be tested in accordance with column 2 of table V, examined for defects listed in table IV, and dimensions specified. Failure to pass any test shall be cause for rejection of the first article. The presence of any defect listed in table IV or any dimension, not with specified requirements, shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 Process examination. Examination shall be made of the following fabrication processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the process and all items processed.

(a) Installation of specified insulation (see 3.4.2 and 3.6.4).

(b) Installation of specified wiring (see 3.5.1).

4.4.2.2 Intermediate testing of insulated wire. To determine compliance with 3.5.1, the wire insulation test specified in 4.6.4 shall be made on a section of each roll of insulated wire used in the manufacture of the ranges. Any sample failing to meet the requirements of 3.5.1 shall be cause for rejection of the entire roll of insulated wire.

4.4.3 End item inspection. The inspection lot shall be the number of ranges of one grade, type, and style offered for inspection at one time.

4.4.3.1 Visual examination. The range shall be examined for defects in table IV. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.



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TABLE IV. Classification of defects

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Finish | Type of finish not as specified (for stainless steel, compare with standard samples available from steel suppliers) | X | |
| | Sharp edges, burrs, or slivers | X | |
| | Top heating surface pitted, porous, cracked, dented, scratched, chipped, rusted | X | |
| | Flakes, blisters, area of no film | | X |
| Construction design and workmanship (general) | Part missing, fractured, split, dented, punctured, bowed, deteriorated, malformed | X | |
| | Design not suitable for battery alignment | X | |
| | Any adjustable part or assembly that cannot be adjusted | X | |
| | Any adjustable part or assembly not properly adjusted | | X |
| | Any part misplaced, loose or not in proper alignment | X | |
| Legs | Leveling devices inoperative | X | |
| Controls | Thermostat or switch not type specified; not properly marked | X | |
| | Thermostat actuating element not protected | X | |
| Indicator lights | Inoperative | X | |
| Top heating surface | Not flat - any visible curvature is unacceptable (check with straight edge) | X | |
| | Back splash not furnished on griddle-top range | X | |
| | Grease receptacles not removeable from front of range | X | |



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TABLE IV. Classification of defects (cont'd)

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Oven | Deck not removable through oven door without removing parts of door | X | |
| | Dampered vent or slide rack not furnished | X | |
| | Heating elements not as specified | X | |
| | Opening not full width or height of oven interior | X | |
| Door | Means for lubricating door hinges not provided, when not permanently lubricated type | X | |
| | Does not remain closed | | X |
| | Not counterbalanced | X | |
| | Does not rest level and even with oven deck | X | |
| | Handles loose, misaligned | | X |
| Broiler | Grid adjustment mechanism defective | X | |
| | Drip pan not of specified material, or does not fit | X | |
| Identification marking and instruction plate | Missing, not of specified material, not marked as specified, illegible, incomplete | | X |
| Materials | Any part or component not fabricated of the applicable referenced materials. (Where not covered by referenced specification.) | X | |

4.4.3.2 Dimensional examination. The range shall be examined for compliance with dimensions specified. Any dimension not within the specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 6.5 total defects, expressed in terms of defects per hundred units.



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* 4.4.3.3 End item testing. When a first article is not required the first produced range shall be tested in accordance with column 4 of table V. Any nonconformance shall be cause for rejection. Tests specified in 4.6.1, 4.6.2, 4.6.6 and 4.6.9, shall be performed on the production items using an inspection level of S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

* 4.4.4 Inspection of preparation for delivery. Inspection shall be made to determine that packaging, packing, and marking are in accordance with MIL-R-11337 as required by section 5 of this specification.

* 4.5 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

* 4.6 Test methods. The test schedule shall be in accordance with table V.

TABLE V. Test schedule

| Col. 1 | Col. 2 | Col. 3 | Col. 4 | | | |
|---|---|---|---|---|---|---|
| Test sequence | First article | Quality conformance | First produced item | Test | Test paragraph | Requirement paragraph |
| 1 | X | X | X | Thermostat test | 4.6.1 | 3.5.1 |
| 2 | X | X | X | Drag test | 4.6.2 | 3.5 |
| 3 | X | | X | Top heating surface time test | 4.6.4 | 3.6.3.1 |
| 4 | X | | X | Temp. distribution test | 4.6.5 | 3.6.3.1 |
| 5 | X | X | X | Door load test | 4.6.6 | 3.6.4.1 |
| 6 | X | | X | Oven heat loss test | 4.6.7 | 3.6.4.3 |
| 7 | X | | X | Oven heating time test | 4.6.8 | 3.6.4.5 |
| 8 | X | X | X | Wattage test | 4.6.9 | 3.5.3 |

4.6.1 Thermostat test. The range shall be connected to an electrical source of nominal voltage and frequency as specified in 3.5.1. The oven temperatures shall be measured by a recording thermocouple suspended from or supported by an oven rack, so that the junction is at least 2 inches from any metal part, is approximately in the center of the usable baking space, and is shielded from direct radiation. Temperature of the top heating surface shall be measured by a laboratory thermometer or thermocouple at approximately the center of the top heating surface. If a thermometer is used,



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the bulb shall be in contact with the top heating surface and shall be well covered with sand. If a thermocouple is used, the hot junction shall be brazed. With the applicable thermostat set at its calibration point, the oven (later, the top heating surface) shall be operated until steady heat condition is reached. Steady heat condition is the condition reached after the heating elements have been under the control of the thermostat for at least the first five cycles of operation of the thermostat, as shown by the indicator light. When this condition is reached, the maximum and minimum oven and top heating surface temperatures shall be taken for three consecutive cycles and the average temperature calculated for compliance with 3.5.4. The oven temperature amplitude per cycle, and the top plate temperature amplitude per cycle, shall be noted. Any noncompliance with the requirements of 3.5.4 shall constitute failure of this test.

4.6.2 Drag test. The completely assembled range shall be dragged once over a concrete floor a distance of 3 feet by the front, back, and both sides to determine compliance with 3.5. Any noncompliance with the strength requirements of 3.5 shall constitute failure of this test.

4.6.3 Wire test. The silicon impregnated asbestos covered wire shall be wound spirally on a 1/2-inch diameter metal mandrel and heated to 550 F for one hour. A 2,500-volt, 60-hz, alternating current potential shall be applied for 1-minute between conductor and mandrel to determine compliance with 3.5.1. Any nonconformance with the silicone-impregnated-asbestos wire-insulation requirements of 3.5.1 shall constitute failure of this test.

4.6.4 Top heating surface time test. With thermometers or thermocouples placed as specified in 4.6.1, each top heating surface shall be energized at rated high heat voltage. The ambient temperature shall be 75 F, plus or minus 5 F. Temperature readings shall be recorded at the beginning and every 5 minutes thereafter up to 20 minutes for the type I and type III ranges, and up to 25 minutes for the type II range, to determine compliance with the heat-up requirements of 3.6.3.1, 3.6.3.2, or 3.6.3.3, as applicable. Any noncompliance with heat-up requirements shall constitute failure of this test.

4.6.5 Temperature distribution test. After 1 hour of heating the top heating surface at the high heat setting, the temperature shall be measured simultaneously at 12 points with location of thermocouples as follows:

(a) One located 1-inch away from each corner, total of four.
(b) One located at the point of intersection of the diagonals.



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(c) The remaining seven shall be located at random with no more than two in one quadrant.

Method of averaging temperature readings shall be as follows:

(d) The average of the peaks shall be obtained by averaging all the instantaneous values at the period of time when the majority were passing through their maximum value.
(e) Similarly, the above method shall be repeated for the valleys.
(f) The maximum average as determined in (d) above shall have a plus or minus 14 percent spread of temperature allowable at the value.
(g) The average surface temperature shall be the average of the maximum and minimum averages as determined in (d) and (e) above and shall be within limits specified in 3.6.3.1, 3.6.3.2 or 3.6.3.3 for even heat distribution.

Any noncompliance with heat distribution requirements shall constitute failure of this test.

4.6.6 Door load test. A dead weight load of 150 pounds shall be placed over the outer edge of the door to determine compliance with door strength requirements of 3.6.4.1. Any noncompliance with 3.6.4.1 shall constitute failure of this test.

4.6.7 Oven heat loss test. A thermocouple shall be placed in the oven as specified in 4.6.1. The oven shall be prepared with all standard baking equipment, such as heating elements, thermostats, deck, and racks in place. A watt-hour meter and watt-meter shall be connected in the circuit. The oven shall be operated for 2 hours at rated-input voltage. The oven thermostat shall be adjusted to maintain a steady internal temperature of 325 F, plus or minus 5 F, above room temperature as indicated by the thermocouple. The test shall be continued for at least 4 hours after the 2 hour conditioning period. The room temperature shall be read every 15 minutes. The steady internal temperature shall be determined every hour and the watt-hour meter shall be read at the beginning and end of the period during which the temperature readings are being taken. The exact time of each reading shall be recorded. The average watts required to maintain steady internal temperature shall be computed $\frac{\text{(watt-hours)}}{\text{hours}}$. The average watts per square foot of inside oven surface shall be computed for compliance with 3.6.4.3. For the purpose of this computation, the oven dimensions shall be:

Height: The maximum vertical interior lining dimension to the nearest 1/4 inch.



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Depth: The horizontal dimension to the closest 1/4 inch, from the inside of the back wall lining to the inside surface of the door, with door completely closed.

Width: The horizontal interior dimension to the closest 1/4 inch, from inside wall to inside wall of oven lining.

Any noncompliance with 3.6.4.3 shall constitute failure of this test.

4.6.8 Oven heating-time test. This test shall be started with the thermocouple placed as specified in 4.6.1, and all parts of the oven at an ambient of 75 F, plus or minus 5 F. Applied rated voltage shall be held within 1 percent throughout the test. The temperature control shall be set at 500 F. Readings of temperature and input voltage shall be recorded at 1-minute intervals; until temperature reaches 450 F. A graph shall be made of temperature versus time to determine time at which the oven temperature reached 325 F above ambient temperature. The time shall be noted to determine compliance with 3.6.4.5. Any noncompliance with 3.6.4 shall constitute failure of this test.

4.6.9 Wattage test. With range connected to a specified electrical power source, the power imput to each heating element shall be measured for compliance with the design power ratings specified in 3.5.3. Any noncompliance with 3.5.3 shall constitute failure of this test.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Each complete range sh be preserved, packaged, packed and marked in accordance with the applicable requirements of MIL-R-11337. Preservation and packaging shall be level A or C, and packing shall be level A, B, or C (see 6.2).

## 6. NOTES

6.1 Intended use. The ranges covered by this specification are intended for use in dining halls for oven roasting and top surface cooking.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, style, and grade of range required (see 1.2).



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(c) When a first article is required (see 3.1, 4.3 and 6.3).
(d) Electrical characteristics, if other than specified (see 3.5.1).
(e) Level of preservation and packaging, and level of packing required (see 5.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.5 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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